

Quiz 1

Website Project Report

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2026

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1 Design Concept and Structure

1.1 Design Concept

The design of this website is inspired by Chakra UI. I wanted to create a minimalist website that is pleasant to look at, easy to navigate, and comfortable to read. This inspiration is reflected in rounded components, subtle borders, consistent spacing, and a clear visual hierarchy. Chakra UI serves as a visual reference for the project. The website is implemented using HTML, Tailwind CSS, custom CSS, and JavaScript with jQuery. Bootstrap Icons provides the icons used for social links, location information, and other interface elements. The website combines a personal portfolio with an introduction to Surabaya, including its local food and tourist destinations. All pages share the same visual style to create a consistent experience.

1.1.1 Minimalist Appearance

The interface uses generous whitespace, restrained colors, and simple components to keep attention on the content. Photographs provide visual interest, while rounded corners and thin borders separate components without making the layout feel crowded. Interactive elements provide subtle feedback. Navigation links receive an accent-colored background when active or hovered. Cards change their border color on hover, while buttons and social links use short transitions. These effects make interactive elements recognizable while preserving the minimalist appearance.

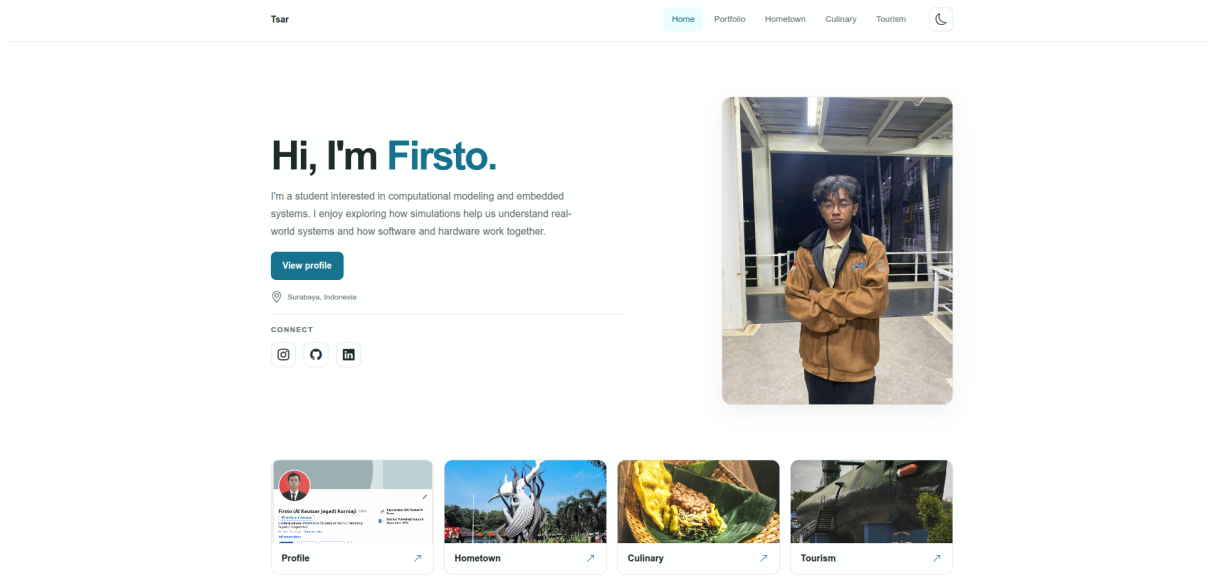


Figure 1: Homepage in light mode.

1.1.2 Typography and Visual Hierarchy

The website uses a sans-serif font through Tailwind's `font-sans` class. This supports the clean, modern appearance of the interface. Headings are larger and heavier than body text, allowing visitors to identify the main sections quickly. Descriptions use smaller text with comfortable line spacing. On the profile page, accent-colored dates, prominent titles, and muted organization

names establish a clear reading order within each timeline entry. Content is placed inside centered containers with a maximum width. This prevents the layout from stretching excessively on large screens and maintains consistent alignment across pages.

1.1.3 Color Palette

The website supports light and dark themes. Colors are defined using CSS custom properties in `assets/css/base.css`, with separate roles for backgrounds, text, borders, and accents. The light theme uses a white background, dark green-toned text, and a cyan-blue accent. Soft neutral colors provide supporting surfaces and borders. The dark theme uses deep green-toned backgrounds, light text, and a brighter cyan accent. Both themes preserve the same color roles to maintain a consistent visual hierarchy.

Color Role	Light Theme	Dark Theme
Page background	#FFFFFF	#111A18
Supporting surface	#F8FAF9	#182420
Primary text	#172724	#E7F1ED
Secondary text	#64736F	#A5B8AE
Border	#E3EAE7	#2E4037
Accent	#0E7490	#67E8F9
Soft accent background	#ECFEFF	#164E63

Table 1: Color palette used.

The accent color highlights the author's name on the homepage, active navigation items, timeline markers, and destination links. Muted text distinguishes supporting descriptions from headings, while subtle borders separate components. The theme button allows visitors to switch between light and dark modes. The selected theme is saved in `localStorage`, allowing the preference to persist across pages and future visits when browser storage is available.

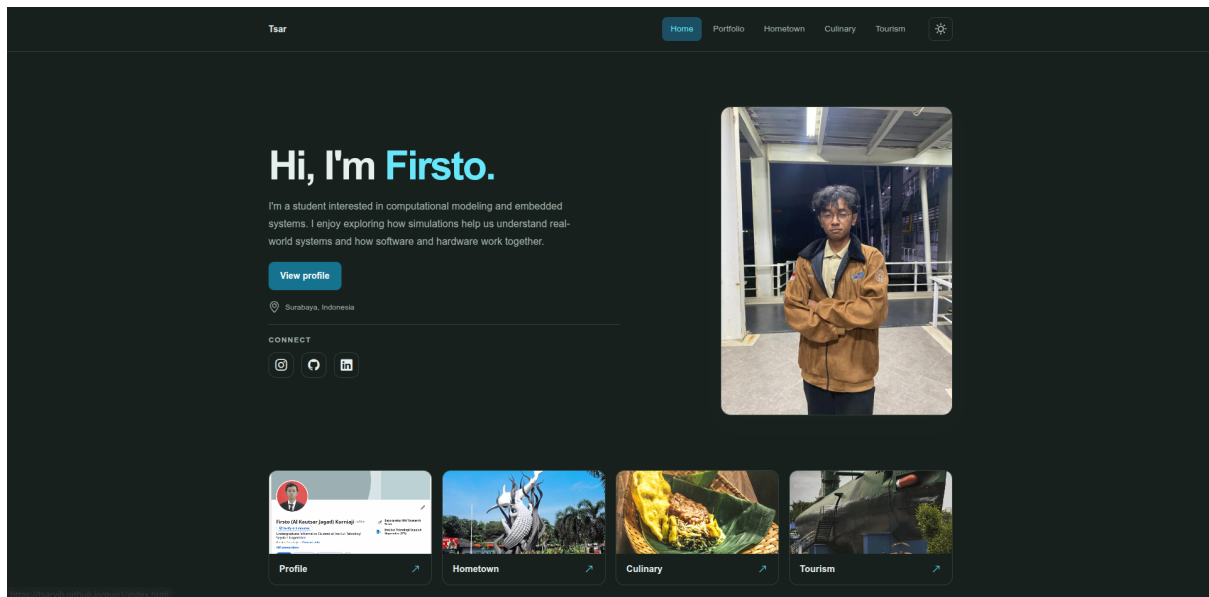


Figure 2: Homepage in dark mode

1.2 Website Structure

The project is organized as a static website with five main pages. Each page has a distinct purpose, while shared navigation allows visitors to move between them.

1. **Home** - `quiz1/index.html`

Introduces the author through a short biography, portrait, location, and social links. Four image cards provide direct access to the other pages.

2. **Profile** - `quiz1/profile/index.html`

Presents personal information, achievements, and experience. On wider screens, a profile sidebar appears beside the timeline content.

3. **Hometown** - `quiz1/hometown/index.html`

Introduces Surabaya through descriptive text, a photograph, and an embedded Google Maps view.

4. **Culinary** - `quiz1/food/index.html`

Displays six local dishes in a card grid: Rawon, Rujak Cingur, Lontong Balap, Sate Klopo, Tahu Tek, and Semanggi Surabaya. Each card contains a photograph, a name, and a short description.

5. **Tourism** - `quiz1/tourist/index.html`

Displays six destinations: Monumen Kapal Selam, Kota Lama, Jalan Tunjungan, Taman Flora, Surabaya North Quay, and Kebun Binatang Surabaya. Each card includes a photograph, location information, and a link to Google Maps.

The shared `assets` directory contains stylesheets, JavaScript files, images, and locally stored libraries. Common styles are defined in `base.css`, while separate stylesheets handle the home-page, card grids, profile layout, and hometown layout. This organization keeps shared design rules consistent while allowing each page to have its own layout.

1.3 Navigation and Responsive Layout

A sticky header keeps the main navigation accessible while visitors scroll. On wider screens, navigation links appear beside the website name and theme button. On smaller screens, a separate navigation row allows the links to wrap across multiple lines, keeping the page links visible without requiring a collapsible menu. The homepage introduction and portrait appear side by side on wider screens and stack on smaller screens. Its navigation cards use two columns on smaller screens and four columns on wider screens. The culinary and tourism grids expand from one column to two and then three columns as the screen becomes wider. The profile and hometown pages similarly change from stacked layouts to two-column arrangements.

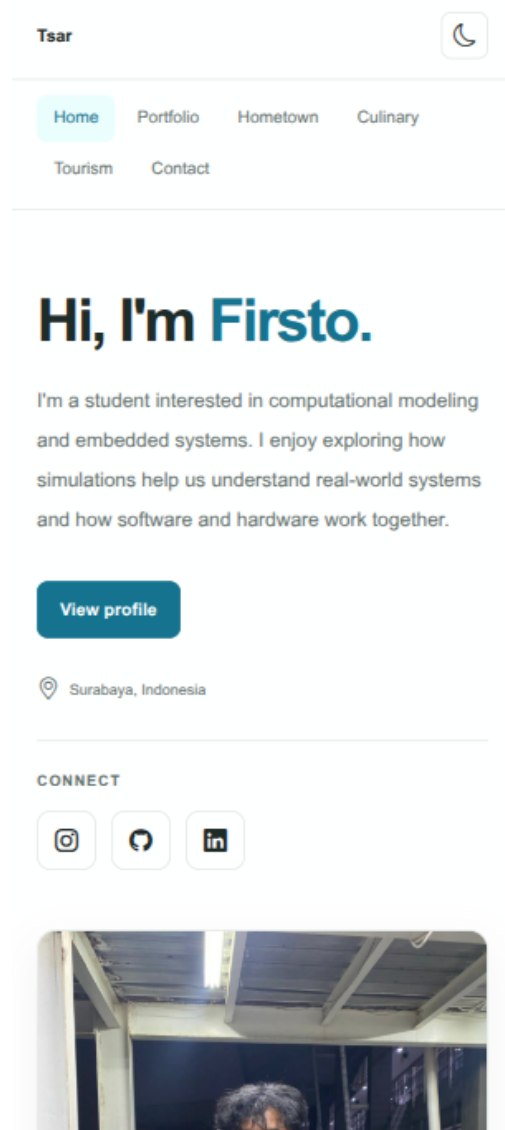


Figure 3: Homepage at a mobile screen width

1.4 Complete Design

1.4.1 Home Page

The homepage introduces the author through a short biography and portrait. Social links and image cards provide access to personal accounts and the other pages.

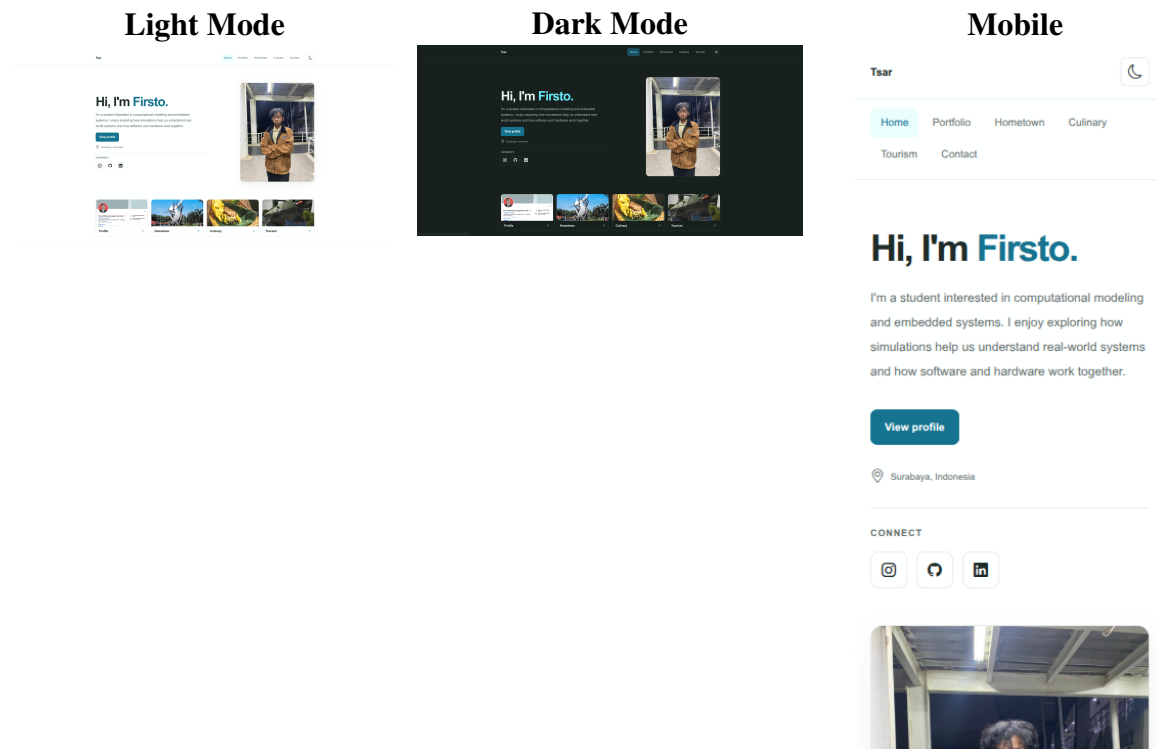


Figure 4: Home page in light mode, dark mode, and the mobile layout.

1.4.2 Profile Page

The profile page combines a personal information card with achievement and experience timelines. These elements appear side by side on wider screens and stack vertically on smaller screens.

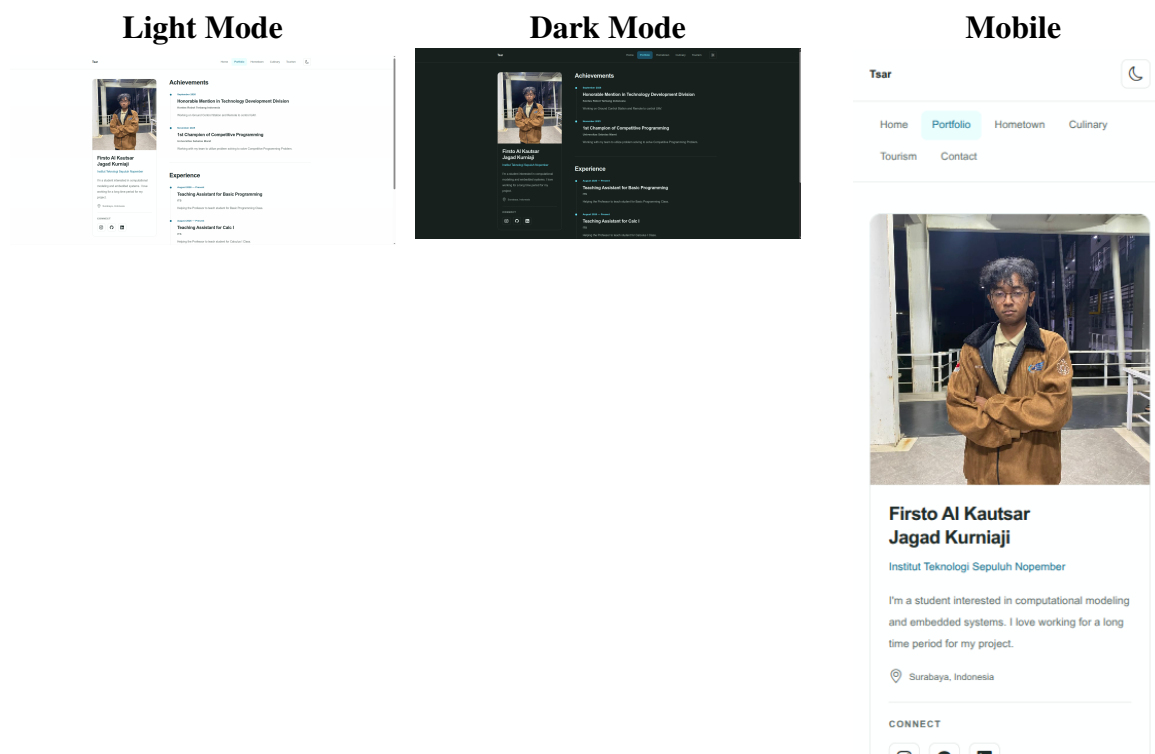


Figure 5: Profile page in light mode, dark mode, and the mobile layout.

1.4.3 Hometown Page

The hometown page introduces Surabaya through descriptive text, a photograph, and an embedded map. Its two-column desktop layout becomes a stacked layout on mobile screens.

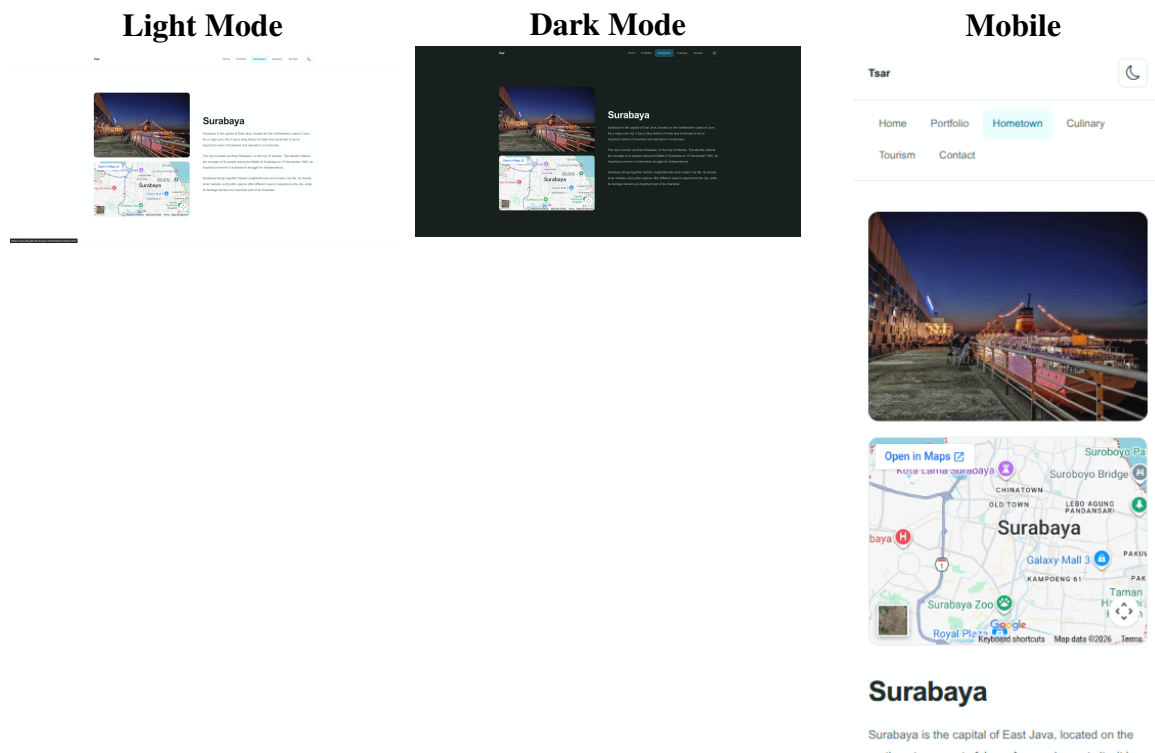


Figure 6: Hometown page in light mode, dark mode, and the mobile layout.

1.4.4 Culinary Page

The culinary page presents six local dishes using image cards. The responsive grid adjusts its column count to fit the available screen width.

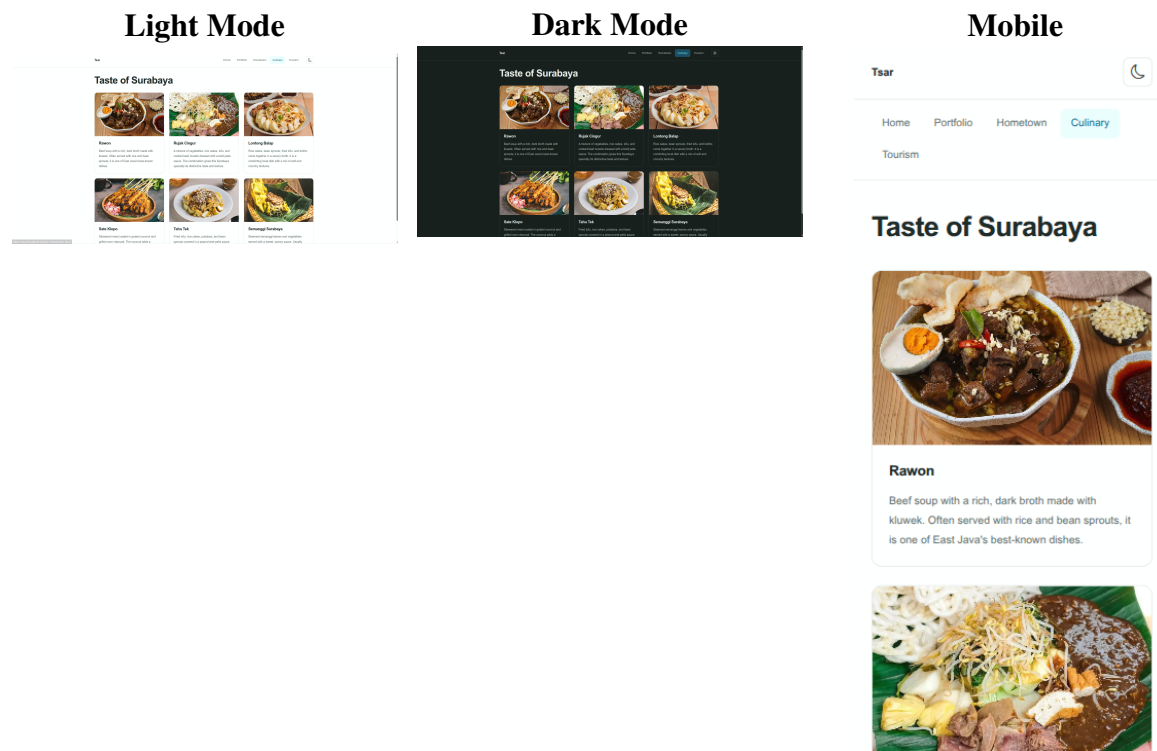


Figure 7: Culinary page in light mode, dark mode, and the mobile layout.

1.4.5 Tourism Page

The tourism page presents six destinations through photographs, location information, and Google Maps links. It uses the same responsive card grid as the culinary page.

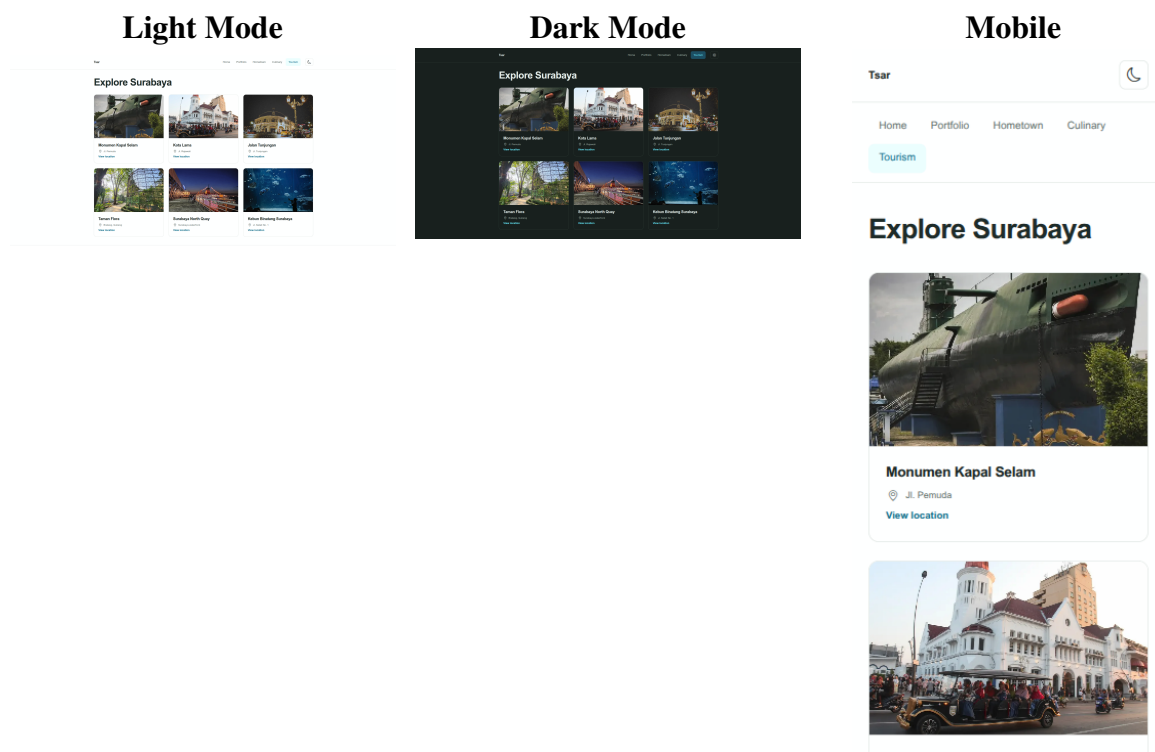


Figure 8: Tourism page in light mode, dark mode, and the mobile layout.

2 Implementation

2.1 Technologies Used

The project is implemented as a static, multi-page website. Each page is an HTML document, while shared stylesheets and scripts provide consistent styling and interactive behavior. The website does not require a backend server or database. The implementation uses the following technologies:

- **HTML5** defines the content, page structure, navigation, images, and embedded map.
- **Tailwind CSS** provides utility classes for spacing, typography, alignment, and responsive layouts.
- **Custom CSS** defines the color themes and page-specific components.
- **JavaScript and jQuery** implement theme switching and synchronize the theme button.
- **Bootstrap Icons** provides social media, location, and theme icons.

The libraries are stored locally in the project. This allows the website to load its styles, scripts, and icons without relying on a third-party CDN. External services are still used for Google Maps and social media links.

2.2 Project Organization

The main pages are located in the quiz1 directory. Shared resources are placed in the assets directory so that multiple pages can use the same files.

```
quiz1/  
  index.html  
  profile/index.html  
  hometown/index.html  
  food/index.html  
  tourist/index.html  
  
assets/  
  css/  
    base.css  
    hero.css  
    nav-card.css  
    card-grid.css  
  pages/  
    profile.css  
    hometown.css  
  js/  
    theme.js  
    site.js  
  img/  
  library/  
    tailwindcss/  
    jquery/  
    bootstrap-icons/
```

The stylesheet `base.css` contains shared rules, including colors, navigation links, buttons, and theme controls. Other stylesheets handle specific components or layouts. For example, `card-grid.css` is shared by the culinary and tourism pages because both display collections of image cards.

2.3 HTML Structure and Navigation

Each page includes a document head, a shared header, and a main content area. The head declares the character encoding, viewport settings, page title, stylesheets, and scripts.

```
<meta charset="utf-8">
<meta name="viewport"
      content="width=device-width, initial-scale=1">
```

UTF-8 supports the characters used in the page content. The viewport declaration allows the responsive layout to follow the device width. Semantic elements organize the content. The header contains the website identity and navigation, while the main element contains the page-specific information. Sections group related content, and article elements represent individual food or tourism cards. Navigation uses ordinary HTML links. Each link points to another HTML document, allowing visitors to move between pages without additional routing logic. The current page is marked explicitly:

```
<a class="nav-link"
    href="../food/index.html"
    aria-current="page">Culinary</a>
```

The `aria-current` attribute identifies the current page, while a shared CSS selector gives that link an accent-colored appearance:

```
.nav-link:hover,
.nav-link[aria-current="page"] {
  color: var(--accent);
  background: var(--accent-soft);
}
```

The header uses Tailwind classes such as `sticky`, `top-0`, and `z-30` to remain visible above the content during scrolling.

2.4 Styling and Reusable Components

Tailwind utility classes handle common layout requirements directly in the HTML. For example, the following classes create a centered content container with a maximum width and horizontal padding:

```
<main id="main"
      class="mx-auto max-w-6xl px-6 py-10 lg:px-8">
```

Custom CSS handles styles that are reused across components or require more specific behavior. Examples include the color palette, destination cards, profile timelines, and portrait sizing. Colors are defined as CSS custom properties:

```
:root {
  --canvas: #fff;
  --surface: #f8faf9;
  --ink: #172724;
```

```

--muted: #64736f;
--border: #e3eae7;
--accent: #0e7490;
--accent-soft: #ecfeff;
}

body {
  background: var(--canvas);
  color: var(--ink);
}

```

This approach gives each color a semantic role. Components reference the role instead of repeating a hexadecimal value, making the palette easier to maintain and allowing theme changes to apply throughout the website. Shared classes such as `destination-card`, `destination-photo`, and `destination-content` keep the culinary and tourism cards visually consistent. Rounded corners, thin borders, and restrained hover effects implement the minimalist design described in Section 1.

2.5 Responsive Layout Implementation

Responsive behavior is implemented through Tailwind breakpoint classes and custom CSS media queries. Layouts change according to the available screen width.

The culinary and tourism pages use CSS Grid. Their shared grid begins with one column, expands to two columns at 40rem, and expands to three columns at 64rem.

```

.destination-grid {
  display: grid;
  grid-template-columns: minmax(0, 1fr);
  gap: 1.5rem;
}

@media (min-width: 40rem) {
  .destination-grid {
    grid-template-columns: repeat(2, minmax(0, 1fr));
  }
}

@media (min-width: 64rem) {
  .destination-grid {
    grid-template-columns: repeat(3, minmax(0, 1fr));
  }
}

```

The homepage navigation cards use two columns on smaller screens and four columns from 48rem. The profile and hometown layouts also switch to two columns from 48rem.

The navigation has separate desktop and mobile arrangements. Tailwind's `hidden md:flex` classes display the desktop links on wider screens, while `md:hidden` displays the wrapping mobile navigation below the main header on smaller screens.

On the profile page, the sidebar becomes sticky on wider screens. Its height is limited relative to the viewport, and vertical scrolling is enabled when its content exceeds the available space.

2.6 Light and Dark Theme Implementation

Theme handling is divided between `theme.js` and `site.js`. The first script restores the saved theme before the stylesheets load. The second initializes the button and responds to user interaction.

2.6.1 Initial Theme Selection

The script `theme.js` reads the saved preference from `localStorage` and applies it to the document's `data-theme` attribute.

```
(function () {  
  try {  
    document.documentElement.dataset.theme =  
      localStorage.getItem("portfolio-theme") || "light";  
  } catch (_) {  
    document.documentElement.dataset.theme = "light";  
  }  
})();
```

If no preference exists, the website uses light mode. The exception handler also selects light mode if browser storage cannot be accessed. Applying the theme early helps reduce a visible change from the default theme during loading.

2.6.2 Theme Switching and Persistence

The script `site.js` runs inside jQuery's document-ready callback. It registers a click handler for the theme button:

```
$("#theme-toggle")  
  .prop("hidden", false)  
  .on("click", function () {  
    const theme =  
      document.documentElement.dataset.theme === "dark"  
        ? "light"  
        : "dark";  
  
    document.documentElement.dataset.theme = theme;  
  
    try {  
      localStorage.setItem("portfolio-theme", theme);  
    } catch (_) {}  
  
    syncTheme();  
  });
```

Changing the attribute activates the corresponding CSS palette. The selected value is saved under the key `portfolio-theme`, which is shared by the pages. The `syncTheme()` function updates the button's `aria-pressed` state and switches between the moon and sun icons. The button is initially hidden in the HTML and is revealed when its JavaScript behavior is initialized. The jQuery library and `site.js` use the `defer` attribute. This allows the browser to parse the HTML while downloading the scripts and preserves their execution order.

2.7 Page-Specific Content Implementation

2.7.1 Homepage

The homepage uses a hero layout containing an introduction, a profile link, social links, and a portrait. Tailwind classes create the wider-screen arrangement, while `hero.css` controls portrait sizing and adjustments for different viewport sizes. Four image cards link to the profile, hometown, culinary, and tourism pages. Their shared styling is defined in `nav-card.css`.

2.7.2 Profile Page

The profile page uses an aside element for the personal summary and separate sections for achievements and experience. Ordered lists organize the timeline entries. Each entry contains a date, title, organization, and description. A left border creates the timeline line, while the CSS pseudo-element `::before` creates an accent-colored marker:

```
.timeline-item::before {
  content: "";
  position: absolute;
  top: .35rem;
  left: -.32rem;
  width: .6rem;
  height: .6rem;
  border-radius: 50%;
  background: var(--accent);
  box-shadow: 0 0 0 .25rem var(--canvas);
}
```

Creating the markers with CSS avoids the need for separate image assets and allows their colors to follow the selected theme.

2.7.3 Hometown Page

The hometown page combines a photograph, descriptive paragraphs, and a Google Maps iframe. The map uses a Surabaya location query and includes a title describing its content. The iframe uses `loading="lazy"` to allow deferred loading. Its width follows the containing column, while custom CSS sets its height, border, and rounded corners.

2.7.4 Culinary and Tourism Pages

Both pages use the shared destination-card structure. Culinary cards include descriptions of local dishes, while tourism cards include location information and links to Google Maps. The content is written directly in HTML. This keeps the implementation straightforward for a small static website, although adding or changing entries requires editing the corresponding HTML document.

2.8 Images, Icons, and External Links

Images are stored in `assets/img` and referenced using relative paths. The culinary and tourism collections use WebP images, while the portrait uses a JPEG image. Destination photographs use a consistent aspect ratio and `object-fit: cover`:

```
.destination-photo {  
  display: block;  
  width: 100%;  
  aspect-ratio: 8 / 5;  
  height: auto;  
  object-fit: cover;  
}
```

This produces consistently shaped cards even when the original images have different dimensions. Cropping allows each photograph to fill its allocated area. Food and tourism images use `loading="lazy"` so that the browser can defer loading images that are not immediately needed. Image elements also declare width and height values to help the browser reserve space during loading. Bootstrap Icons supplies recognizable symbols through classes such as `bi-github`, `bi-geo-alt`, `bi-moon`, and `bi-sun`. Using a shared icon library maintains a consistent appearance. Social media links and tourism location links open in a new tab using `target="_blank"`. They also include `rel="noopener noreferrer"`. Tourism links use Google Maps search URLs, allowing visitors to open a destination's location directly from its card.

3 Analysis and Evaluation

3.1 Achievement of Project Objectives

The project implements five connected webpages: Home, Profile, Hometown, Culinary, and Tourism. Together, these pages introduce the author and present information about Surabaya, its food, and its tourist destinations. The implementation reflects the intended minimalist design through consistent spacing, rounded components, subtle borders, and a restrained color palette. Shared navigation and reusable styles give the pages a coherent appearance despite their different content. This evaluation is based on the project's source code and implemented features; it does not represent formal user testing or measured performance results.

3.2 Visual Design and Content Presentation

One strength of the project is its consistent visual hierarchy. Headings identify the main subjects, muted descriptions provide supporting information, and accent colors highlight active navigation items, timeline dates, and interactive links. The layouts suit their content: the homepage provides a concise introduction, the profile page organizes experience chronologically, the culinary and tourism pages present collections through cards, and the hometown page combines descriptive text with a photograph and map. Light and dark themes preserve the same component structure and color roles, providing visual continuity when changing themes. However, text and interface contrast should be checked systematically before claiming compliance with accessibility standards.

3.3 Navigation and Responsive Behavior

The shared navigation provides direct access to all five pages, while highlighting the current page helps visitors understand their position within the website. Homepage image cards provide another way to explore the content. The source code defines responsive layouts in which card grids change their column count and the homepage, profile, and hometown content move from side-by-side arrangements to stacked layouts. Mobile navigation keeps links visible by allowing them to wrap, although this occupies additional vertical space on narrow screens. Testing on different devices would help determine whether the spacing and wrapping remain comfortable. Full-page mobile screenshots can also become difficult to read when reduced for the report, so detailed screenshots of individual components could supplement the overview images.

3.4 Code Organization and Maintainability

Separating shared styles from page-specific styles makes the project easier to understand. CSS custom properties allow the color palette to be maintained centrally, while shared card classes keep the culinary and tourism pages consistent. The division between `theme.js` and `site.js` also has a clear purpose: the first restores the theme early, while the second manages interaction after the document is ready. A limitation is that the header and navigation markup are repeated across five HTML files, requiring future changes to be applied consistently to each copy. Food, destination, and timeline entries are also written directly in HTML. This approach is manageable for the current project size, but reusable templates and structured data could simplify maintenance if the website grows.

3.5 Loading and External Dependencies

The static architecture avoids backend processing and database requests, while locally stored libraries allow the main styles, scripts, and icons to load without a third-party CDN. Lazy loading is used for collection images and the embedded map, potentially reducing the resources requested immediately depending on browser behavior and content position. Explicit image dimensions also help the browser reserve space before images finish loading. However, these choices do not establish the website's actual loading speed, which still depends on image sizes, library sizes, network conditions, and browser behavior. A browser audit would be needed to measure performance and identify improvements. Google Maps remains an external dependency, so the embedded map may not display when the service is unavailable or the visitor has no internet connection. Social media and destination links also require access to external websites.

3.6 Limitations and Possible Improvements

Several details could be improved in a future revision:

- **Profile social links:** The profile page currently links to the general Instagram, GitHub, and LinkedIn websites. These should point to the author's accounts, consistent with the homepage.
- **Portrait description:** The profile portrait does not currently include an `alt` attribute. A meaningful alternative description should be added.
- **Repeated markup:** A shared template could simplify updates to the header and navigation across all pages.
- **Content proofreading:** Some profile descriptions contain grammatical errors or inconsistent wording. Revising them would improve the presentation.
- **Verification:** Cross-browser checks, keyboard navigation checks, contrast measurements, and performance audits would provide stronger evidence of quality.

These improvements would strengthen content consistency, maintainability, and usability while preserving the project's overall design direction.

4 Conclusion

The Quiz 1 project demonstrates the implementation of a static, multi-page website using HTML, Tailwind CSS, custom CSS, and JavaScript with jQuery. Its five pages combine a personal introduction with information about Surabaya, local dishes, and tourist destinations. The Chakra UI-inspired design supports the goal of creating a minimalist website that is pleasant to look at through consistent spacing, reusable components, and a shared color palette. Responsive layouts accommodate different screen widths, while light and dark themes provide an additional presentation option. The project also demonstrates the value of organizing shared resources and defining reusable styling rules. Further improvements should focus on correcting content details, reducing repeated markup, and verifying behavior across browsers and devices. The implementation provides a functional foundation that can be expanded as the author's portfolio and website content grow.